

A Technical Introduction to Speech recognition

Section 1

Conferences Regular conferences include SpeechTEK and SpeechTEK Europe, ICASSP, Interspeech/Eurospeech, and the IEEE ASRU. Conferences in the field of natural language processing, such as ACL, NAACL, EMNLP, and HLT, include papers on speech processing.

Section 2

Automatic pronunciation assessment is the use of speech recognition to verify the correctness of speech, as distinguished from assessment by a person. Also called speech verification, pronunciation evaluation, and pronunciation scoring, the main application of this technology is computer-aided pronunciation teaching (CAPT) when combined with computer-aided instruction for computer-assisted language learning (CALL), speech remediation, or accent reduction. Pronunciation assessment does not determine unknown speech (as in dictation or automatic transcription) but instead, compares speech to a reference model for the words spoken, sometimes with inconsequential prosody such as intonation, pitch, tempo, rhythm, and stress. Pronunciation assessment is also used in reading tutoring, for example in products such as Microsoft Teams and Amira Learning. Pronunciation assessment can also be used to help diagnose and treat speech disorders such as apraxia. Assessing intelligibility is essential for avoiding inaccuracies from accent bias, especially in high-stakes assessments, from words with multiple correct pronunciations, and from phoneme coding errors in digital pronunciation dictionaries. In 2022, researchers found that some newer speech to text systems, based on end-to-end reinforcement learning to map audio signals directly into words, produce word and phrase confidence scores closely correlated with listener intelligibility. In the Common European Framework of Reference for Languages (CEFR) assessment criteria for "overall phonological control", intelligibility outweighs formally correct pronunciation at all levels.

Section 3

Aircraft Substantial efforts have been devoted to the test and evaluation of speech recognition in fighter aircraft. Of particular note have been the US programme in speech recognition for the Advanced Fighter Technology Integration (AFTI)/F-16 aircraft (F-16 VISTA), the programme in France for Mirage aircraft, and UK programmes dealing with a variety of aircraft platforms. In these programmes, speech recognizers have been operated successfully, with applications including setting radio frequencies, commanding an autopilot system,

setting steer-point coordinates and weapons release parameters, and controlling flight display. Working with Swedish pilots flying the JAS-39 Gripen, Englund (2004) reported that recognition deteriorated with increasing g-loads. The study concluded that adaptation greatly improved the results in all cases and that the introduction of models for breathing was shown to improve recognition scores significantly. Contrary to what might have been expected, no effects of the broken English of the speakers were found. Spontaneous speech caused problems for the recognizer, as might have been expected. A restricted vocabulary, and above all, a proper syntax, could thus be expected to improve recognition accuracy substantially. The Eurofighter Typhoon employs a speaker-dependent system, requiring each pilot to create a template. The system is not used for safety-critical or weapon-critical tasks, such as weapon release or lowering of the undercarriage, but is used for many cockpit functions. Voice commands are confirmed by visual and/or aural feedback. The system is seen as a major benefit in the reduction of pilot workload, and allows the pilot to assign targets with two voice commands or to a wingman with only five commands. Speaker-independent systems are under test for the F-35 Lightning II (JSF) and the Alenia Aermacchi M-346 Master lead-in fighter trainer. These systems have produced word accuracy scores in excess of 98%.

Section 4

2010s By early the 2010s, speech recognition was differentiated from speaker recognition, and speaker independence was considered a major breakthrough. Until then, systems required a "training" period for each voice. In 2017, Microsoft researchers reached the human parity milestone of transcribing conversational speech on the widely benchmarked Switchboard task. Multiple deep learning models were used to improve accuracy. The error rate was reported to be as low as 4 professional human transcribers working together on the same benchmark.

Section 5

Models, methods, and algorithms Both acoustic modeling and language modeling are important parts of statistically-based speech recognition algorithms. Hidden Markov models (HMMs) are widely used in many systems. Language modelling is also used in many other natural language processing applications, such as document classification or statistical machine translation.

Section 6

Helicopters The problems of achieving high recognition accuracy under stress and noise are particularly relevant in the helicopter environment as well as in the fighter environment. The acoustic noise problem is actually more severe in the helicopter environment, because of the high

noise levels, and because helicopter pilots, in general, do not wear a facemask, which would reduce acoustic noise in the microphone. Substantial test and evaluation programmes, notably by the U.S. Army Avionics Research and Development Activity (AVRADA) and by the Royal Aerospace Establishment (RAE) in the UK. Work in France included speech recognition in the Puma helicopter. Voice applications include control of communication radios, navigation systems, and an automated target handover system. The overriding issue for voice is the impact on pilot effectiveness. Encouraging results are reported for the AVRADA tests, although these represent only a feasibility demonstration in a test environment. Much remains to be done both in speech recognition and in overall speech technology in order to consistently achieve performance improvements in operational settings.

Section 7

Acoustic signals are structured into a hierarchy of units, e.g. phonemes, words, phrases, and sentences; Each level provides additional constraints; e.g., known word pronunciations or legal word sequences, which can compensate for errors or uncertainties at a lower level; This hierarchy of constraints improves accuracy. By combining decisions probabilistically at all lower levels, and making ultimate decisions only at the highest level, speech recognition is broken into several phases. Computationally, it is a problem in which a sound pattern has to be recognized or classified into a category that represents a meaning to a human. Every acoustic signal can be broken into smaller sub-signals. As the more complex sound signal is divided, different levels are created, where at the top level are complex sounds made of simpler sounds on the lower level, etc. At the lowest level, simple and more probabilistic rules apply. These sounds are put together into more complex sounds on upper level, a new set of more deterministic rules predicts what the complex sound represents. The upper level of a deterministic rule should figure out the meaning of complex expressions. In order to expand our knowledge about speech recognition, we need to take into consideration neural networks. Neural network approaches use the following steps: